# AUtoMLy | Medicaid Policy Copilot

## INAI 6986 — Milestone 2: Model Testing, Validation and Refinement

J.O. • Dr. David M. Simmonds • October 18, 2026

**Status: development results; the one-time final test has not been run**

The final test needs a fresh, locked benchmark that does not exist yet (its protocol is declared in Section 2). Every result below comes from held-out validation questions the models were never trained on, and each table says so. I make no final verdict and no deployment claim.

## 1. Summary of Results

The baseline is Nemotron 3.5 Lightning's untouched base model; the model is the same base with my fine-tuned adapter (1,600 training records, checkpoint 400). My PM Plan threshold is at least 10 percentage points more correct, fully supported answers than the base, with fewer than 2% of answers containing an unsupported claim. On the 24-question development probe both gave 4 fully supported answers (17 per 100 questions): a difference of 0 points, so the threshold is not met. On the 15 questions I reviewed, the adapter gave 4 strict answers and the base 2 (27 vs. 13 per 100), a 13-point difference that is not statistically reliable ($p = 0.5$). The adapter put an unsupported claim in 19 of 24 answers (79 per 100) and never declined. In business terms, an analyst would get about 17 usable answers per 100 questions and would have to catch about 79 answers with an invented or wrong policy detail. Verdict on development evidence: the model does not meet the threshold and would not be deployed. The final test will confirm or overturn this.

## 2. Experimental Setup

**Frozen configuration.** NVIDIA Nemotron 3.5 Lightning 30B-A3B, BF16, revision a9904d2. Adapter: LoRA rank 16 trained on 1,600 records, checkpoint 400 (adapter SHA-256 4aaca5c5…). Decoding: greedy (no sampling), at most 512 new tokens, reasoning mode off, seed 20261006. The base condition is the same weights with the adapter switched off. Development runs gave the model only the question. On October 14 I changed the final-test input: both conditions receive the same official source passage with the question, so the test measures whether fine-tuning improves reading and citing the right rule rather than recalling a manual from memory. No retrieval system is involved.

**Split and test protection.** Records are split by document (1,684 train / 124 validation / 191 test). All tuning used validation records only: checkpoints were chosen by validation loss, and the 24-question probe and the Week 8 review used validation questions. The 191 historical test records were used for diagnostics in September, so I no longer treat them as untouched. The final test will use fresh questions from official documents outside all three splits. An October 14 audit found no exposure evidence for 11 of 12 candidate documents and flagged one pharmacy chapter, which is not used unless its review clears it. The questions are written by a local AI model and checked against the source, not written by me. Membership, prompts, settings and scoring rules are locked before the single run, and nothing is tuned afterward.

**Metrics.** Primary: factual accuracy on answerable questions; an answer counts only if it is correct, supported by the source and free of unsupported claims (strict accuracy). Secondary: unsupported-claim rate, correct abstention on unanswerable questions and citation correctness. They reflect the business asymmetry: a confident wrong policy statement can lead to a wrong billing decision, while a refusal only costs analyst time.

**Changes from the PM Plan, declared October 18 before any final output exists.** (1) The PM Plan committed to human-judged accuracy. The final test keeps that human check but changes how it is done: two LLM judges, OpenAI GPT Luna and Anthropic Claude Sonnet 5.5, first score every output, and I then review both judges' ratings for every output. Where the judges disagree, I decide. Where I find an error in a rating, I record it and my correction becomes the final rating; otherwise the judges' rating stands. This validates every output by hand instead of a sample of about 36. An answer passes only if, after my review, it is correct, supported by the source and free of unsupported claims. I report each judge's results, how often the two judges agreed, and how many ratings I changed. Luna replaces the Qwen judge proposed earlier, so neither judge comes from the Qwen family that generated the training questions or the Nemotron family under test; the historical Qwen development scores are unchanged. The judges run separately with the same rubric and source passage and do not see each other's ratings. Model identifiers, prompts and access routes are recorded and locked before the run; Claude runs through the signed-in Claude Code command-line tool (Hermes Direct Claude Code SDK workflow), not the Claude API. (2) The October development judge matched my review on correctness, support and unsupported claims but on citation validity in only 3 of 12 shared cases (Section 5). Code therefore checks each cited document and page against the expected source, and the judges and I check whether that source supports the answer.

**Final-test protocol, locked and fingerprinted before the single run**

| Element | What is fixed in advance |
|---|---|
| Benchmark | Fresh questions written from official documents outside all three splits by a local model and checked against the source page; about 120 answerable and 30 unanswerable questions planned (the accepted set sets the final counts). The flagged pharmacy source is excluded unless its review clears it. |
| Conditions | Nemotron adapter (checkpoint 400) and its untouched base, each given the same source passage and question; greedy decoding, 512 new tokens, seed 20261006. One run; failed outputs stay in the denominator; no retries or tuning afterward. |
| Scoring | OpenAI GPT Luna and Anthropic Claude Sonnet 5.5 score every output; I review every rating, decide disagreements and correct errors. Pass = correct, supported, no unsupported claim after my review. Citation identifiers are checked in code; on unanswerable questions a correct answer is a decline. |
| Robustness | Each passage cut to its first 60% of characters; questions whose evidence survives the cut and questions whose evidence is removed are scored separately. |
| Subgroups | Document family, and answerable versus unanswerable questions. |
| Thresholds | From the PM Plan: +10 points over base on answerable accuracy; at least 95% correct citations; under 2% unsupported claims; at least 90% correct abstention. Safety routing and general-task regression are reported as not tested unless the locked protocol adds them. |

## 3. Iteration Narrative

The run log (16 training runs and four development experiments) gives this arc. Failed attempts stay in the record because they ruled things out.

| When | What I tried | Expected | What happened | What I concluded |
|---|---|---|---|---|
| Sep 16–18 | First Muse fine-tunes | Adapter beats base | Used earlier data-generation and training methods that I later amended | Excluded the runs; all later runs use the amended data, split and procedure. |
| Sep 29–Oct 1 | 16 runs: 4 families × 200 / 509 / 750 / 1,600 records, same validation set | More data lowers validation loss | True for three families; Muse flat after 509. From 509 records up, every family overfit after 1–2 epochs. Early Nemotron and Ornith attempts failed (one on an unsupported warm-up setting) | Use 1,600 records and the early checkpoint. Fixed the training runner and reran all eight runs from scratch. |
| Oct 6–7 | Week 8 error review: 15 questions × base and adapter (Nemotron) | Adapter more accurate | Strict accuracy 2/15 → 4/15; 0 of 30 citations correct; invented rules most common | Citations are format only; question-only input invites recall errors. |
| Oct 7 | Refinement: added an instruction to use only verifiable source facts (8 questions, 2 documents) | Fewer unsupported claims | Base swapped answers for “cannot verify” (correct 3/8 → 0/8); adapter unchanged at 4/8 and still cited wrong sources on 8/8 | Prompt wording alone does not fix grounding. Not promoted. |
| Oct 8 | Cross-family probe: 192 outputs scored by a local AI judge | A fair comparison | The first judge counted refusals as supported answers, so I withdrew it; the corrected judge requires a real answer | Three adapters tie at 4/24; selected Nemotron (Milestone 1). |
| Oct 14 | Changed the final-test input to supply the source passage | Isolates what fine-tuning adds | Evaluation harness built; passed separate specification and code-quality reviews | Ready once the fresh benchmark is locked. |

What the failures ruled out: the early data methods, longer training (more epochs overfit), prompt-only fixes and refusal-friendly scoring. What remains unproven is whether fine-tuning helps when the source is supplied, which is exactly what the final test measures.

## 4. Test Results (Development; Final Test Pending)

The required single test has not been run, so Table 1 gives the development equivalent: the same 24 validation questions (9 documents), the same settings and metric, base versus model versus threshold. The final table will have the same rows, computed once on the locked benchmark, with answerable and unanswerable questions reported separately so a pooled score cannot hide an abstention failure.

**Table 1. Baseline, model and threshold on the same development questions**

| Measure | Untouched base | Adapter (model) | Threshold | Development verdict |
|---|---|---|---|---|
| Fully supported answers, 24 questions (AI-judged) | 4/24 (17%) | 4/24 (17%) | +10 points over base | Not met (+0) |
| Strict accuracy, 15 questions (my review) | 2/15 (13%) | 4/15 (27%) | +10 points over base | +13 points; not reliable (p = 0.5) |
| Answers with an unsupported claim, 24 questions | 17/24 (71%) | 19/24 (79%) | Under 2% | Not met |
| Citation points to the correct page, 15 questions | 0/15 | 0/15 | At least 95% | Not met |

| Measure | Untouched base | Adapter (model) | Threshold | Development verdict |
|---|---|---|---|---|
| Correct abstention on unanswerable questions | Not measurable | Not measurable | At least 90% | Not tested: no unanswerable development questions |

**Table 2. Confusion-matrix equivalent: outcomes on the 24 answerable development questions**

| Outcome | Untouched base | Adapter |
|---|---|---|
| Answered, fully supported | 4 | 4 |
| Answered, but wrong, partly wrong or unsupported | 17 | 20 |
| Declined to answer (unnecessary refusal) | 3 | 0 |
| Generation failed | 0 | 0 |
| Unanswerable questions: correctly declined / answered unsafely | None in development set | None in development set |

**Business units.** Per 100 analyst questions like these, either version would give about 17 answers an analyst could use as drafted. The adapter would give about 79 answers with an invented or wrong detail (the base about 71) and would never say it cannot answer. Manual lookup takes about 84 seconds per simple question, but reviewing and correcting 79 flawed drafts would cost more than 17 usable drafts save, so the development result supports no time-saving claim.

## 5. Error and Robustness Analysis

The failure modes below come from my Week 8 review and were re-checked on the 24-question probe, both on the frozen candidate.

**Table 3. Named failure modes with counts (one label per output)**

| Failure mode | Week 8 review: base / adapter (15 each) | Probe, AI judge: base / adapter (24 each) | What it suggests |
|---|---|---|---|
| Invented rule: states a rule the source does not contain | 9 / 10 | 11 / 4 | Input and data change: supply the source passage; add unanswerable training examples. |
| Wrong value: wrong number, date or code | 3 / 0 | 6 / 15 | Data change: check effective dates and editions. Scope rule: show the quoted source value beside the answer. |
| Wrong citation on an otherwise correct answer | 3 / 5 | 0 / 2 | Scope rule: insert the citation from the supplied passage instead of letting the model write it. |
| Abstention problem: declined when the answer was available | 0 / 0 | 3 / 0 | Data change: teach when to decline, using unanswerable examples. |
| Missing evidence: answer without the needed support | 0 / 0 | 0 / 1 | Input change: supply the source passage. |
| No error | 0 / 0 | 4 / 2 | — |

The review also marked “failed to decline” on 10 base and 6 adapter answers. The two sources disagree on the label for half the shared outputs (6 of 12), mostly “wrong value” versus “invented rule”, but they agree that nearly every adapter error is a confident wrong statement rather than a refusal. Across all 150 judged probe outputs (all families), the AI judge found 57 wrong values, 32 abstention problems, 26 invented rules, 8 wrong citations and 26 answers with no error.

**Check on the measurement itself.** On 12 Nemotron outputs scored by both the AI judge and my review, they agreed on correctness, evidence support, unsupported claims and the overall pass in 12 of 12 cases, but on citation validity in only 3 of 12. My review was AI-assisted (I saw the judge’s suggestion and confirmed all 180 ratings), so I do not reuse those ratings for the final test. The final test uses two LLM judges whose ratings I review, plus a code check for citations (Section 2).

**Robustness observation: reworded instructions.** On October 7 I reran 8 development questions with a rewritten instruction (use only facts you can verify). For the frozen adapter, correct answers stayed at 4/8 and wrong citations at 8/8, so the measure moved 0 points. The base model was fragile: correct answers fell from 3/8 to 0/8 (−38 points) as it switched to refusing. This was run as a refinement attempt, judged by the AI judge, on only 8 questions, so it is weak evidence; I report it because it shows how far the primary measure moves under a realistic prompt change.

**Predeclared final robustness check (not yet run).** Each final question is also run with the source passage cut to its first 60% of characters, a realistic loss of context. Questions whose evidence survives the cut and questions whose evidence is removed are scored separately. The expected behavior is unchanged accuracy when the evidence survives and a correct refusal when it is removed. This evaluates the frozen model and does not reopen tuning.

## 6. Subgroup Performance

The segments my decision affects are the two document families an analyst consults (Administrative Code rules and Provider Manual billing chapters) and answerable versus unanswerable questions.

**Table 4. Development results by segment (24-question probe, AI-judged)**

| Segment | Base: fully supported | Adapter: fully supported | Adapter: unsupported claims |
|---|---|---|---|
| Administrative Code (12 questions) | 2/12 | 1/12 | 11/12 |
| Provider Manual (12 questions) | 2/12 | 3/12 | 8/12 |
| Unanswerable questions | Not in development set | Not in development set | — |

Gaps as found: the adapter did slightly better on Provider Manual questions and worse on Administrative Code questions, where 11 of 12 answers carried an unsupported claim. With 12 questions per group a two-answer gap is within chance, and Gemma's adapter shows the opposite pattern (3/12 Administrative Code vs. 1/12 Provider Manual), so I read this as noise rather than a stable gap. What I would do: the final benchmark reports each document family with its own denominator, and if Administrative Code accuracy trails, I would limit model drafting to Provider Manual questions. My Week 8 review covered only Provider Manual questions, which is itself a coverage gap.

**NIST AI RMF, tied to these results.** Measure: a 79% unsupported-claim rate fails the framework's "valid and reliable" characteristic, and 0 of 30 correct citations fails "accountable and transparent", because a reviewer cannot trace the answer to its rule. Manage: these results trigger the deployment block in my plan, so the model stays in testing. Govern: an analyst must approve every answer, and a named owner tracks manual editions. Map: the harm is wrong billing guidance reaching providers, which can delay or deny services to Medicaid members.

**EU AI Act.** The system is used in Alabama, so the Act does not apply directly. If it did, a tool that evaluates people's eligibility for public benefits would be high-risk (Annex III). My scope excludes eligibility decisions and keeps a human reviewer, which should keep it outside that category, though transparency duties for AI-generated text could still apply.

## 7. Limitations and Deployment Readiness

- **Population and time validity.** Results cover Alabama Medicaid's public Administrative Code and Provider Manual as downloaded September 14, 2026. The manuals are reissued in dated editions (the corpus holds April and July 2026 files), so an adapter trained on one edition goes stale when the next appears. The development questions came from 9 documents and were all answerable; nothing here speaks to other states or to unanswerable questions.
- **Remaining failure modes.** Invented rules and wrong values, wrong citations and no refusals (Section 5).
- **Why the final test is not done.** The fresh benchmark must be generated, source-checked and locked first (protocol in Section 2), and the one flagged source must be reviewed. With about 30 unanswerable questions, abstention will be reported with its interval rather than as a proven 90% rate. LLM judges can share blind spots, so I review every rating, check citations in code, and report how many ratings I changed.
- **Operational requirements.** One GPU with about 64 GB available for the model (63.2 GB allocated after loading on a 96 GB RTX PRO 6000 Blackwell); median answer time 2.9 seconds (90th percentile 3.4 seconds); the source passage retrieved and passed in with each question; analyst approval before any answer is used; and an audit log of question, passage, answer and reviewer decision.

**Table 5. Proposed monitoring plan for any pilot**

| What I would monitor | How | Trigger and response |
|---|---|---|
| Unsupported-claim rate | An analyst checks a random 25 answers each week | Above 2% two weeks running: stop drafting and re-evaluate |
| Citation correctness | Same weekly sample, checked against the cited page | Below 95%: switch to inserted citations and re-evaluate |
| Abstention | Five seeded unanswerable questions each week | Fewer than 90% declined: block use |
| Source editions | Compare manual edition dates with the training snapshot monthly | New edition of a covered chapter: re-run the evaluation; retrain only if accuracy drops |
| Latency | Log the answer time of every request | 90th percentile above 10 seconds: capacity review |

**Readiness: not ready.** A pilot requires passing every threshold on the final test. Autonomous policy decisions stay out of scope even then.

## 8. Repository

Course repository (public): github.com/porespellar/INAI6986_project_deliverables

Development notebook: notebook/current_evidence_replay_20261018.ipynb

The notebook runs top to bottom offline with zero errors and recomputes every development number in this report from the stored records, after checking each input's fingerprint. It does not rerun model generation. A final-test notebook that reproduces the reported test result from the locked outputs and ratings will follow the one-time test.

## AI Use Disclosure

I used Hermes Agent with GPT-6 Luna and GPT-6.1 Sol (October 1–16) and Claude Opus 5.5, Claude Sonnet 5.5 and GPT-6.1 Sol (October 18) to draft and revise this report and write its evaluation, analysis and build code. Local Qwen models generated training records and historical advisory scores. I designed the evaluation, chose the candidate and thresholds, made the Week 8 ratings, and chose the final-test input and the Luna/Sonnet judging that I will review myself. I checked the reported figures against the notebook output and remain responsible for the content.

## Appendix A — Development Results for All Four Families

Same 24 validation questions, question-only input, greedy decoding, AI-judged. Failed generations count as not correct. Paired exact McNemar tests compare each adapter with its own base.

| Model | Condition | Fully supported (95% CI) | Unsupported claims | Abstained | Failed | Paired p |
|---|---|---|---|---|---|---|
| Nemotron | Untouched base | 4/24 (7%–36%) | 17/24 | 3 | 0 | |
| Nemotron | Adapter | 4/24 (7%–36%) | 19/24 | 0 | 0 | 1.0 |
| Gemma | Untouched base | 0/24 (0%–14%) | 10/24 | 14 | 0 | |
| Gemma | Adapter | 4/24 (7%–36%) | 18/23 | 0 | 1 | 0.125 |
| Ornith | Untouched base | 1/24 (1%–20%) | 0/24 | 23 | 0 | |
| Ornith | Adapter | 4/24 (7%–36%) | 18/23 | 0 | 1 | 0.375 |
| Muse | Untouched base | 0/24 (0%–14%) | 0/8 | 8 | 16 | |
| Muse | Adapter | 0/24 (0%–14%) | — | — | 24 | 1.0 |

## Appendix B — Evidence Files

| Claim | File in the source-of-record repository |
|---|---|
| Predeclared probe: questions, settings, selection rule | artifacts/milestone2_completion/development/shortlist_v1/frozen/contract.json |
| 192 counted outcomes and judgments | artifacts/milestone2_completion/development/shortlist_v1/verification/development_v2_counted_outcomes_20261008.jsonl |
| Week 8 review (my ratings, revision 11) | artifacts/milestone2_completion/review/batch_b/verification/submitted_revision_000011/ |
| Refinement experiment | artifacts/milestone2_completion/development/batch_b/refinement_v1/outputs/ |
| Final-test harness reviews | Runtime code fingerprint b4c06b7f5a85770a…; specification and code-quality reviews dated October 14 |
| Exposure audit of fresh sources | October 14 audit receipt (12 documents; 11 with no exposure evidence; 1 flagged) |
| All derived numbers in this report | artifacts/milestone2_completion/class_package_20261018/development_analysis_20261018.json |